

Canonical Dynamical Geometry System v4 — Gap Closure Addendum

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Status: Fills open problems P5, P6, P7 and five practitioner gaps identified in structural analysis.

Prerequisite: All notation, assumptions (A1–A5), and theorem numbers follow canonical_system_v4_3.pdf verbatim.

G1. Discretisation Error (Open Problem P5)

G1.1 Euler scheme

Fix step size $h > 0$. The explicit Euler discretisation of the canonical ODE is

$$X_{t+h} = X_t + h \left[F_{\mathrm{base}}(X_t) - g_X(X_t) - \gamma(X_t), \frac{\angle F(X_t), g_X(X_t)}{|g_X(X_t)|^2}, g_X(X_t) \right].$$

Proposition G1.1 (Euler energy increment). Under (A1)–(A4) and the angle condition $\angle g_X, F_{\mathrm{base}} \leq |g_X|^2$, the one-step energy increment satisfies

$$E(X_{t+h}) - E(X_t) \leq h \dot{E}(X_t) + \frac{1}{2} h^2 \Lambda(X_t) \|\dot{X}_t\|_{\mathrm{op}}^2,$$

where $\Lambda(X_t) := \lambda_{\max}(\nabla^2_X E(X_t))$ and $\|\cdot\|_{\mathrm{op}}$ is the operator norm of $\nabla^2_X E$.

Proof sketch. Taylor-expand $E(X_{t+h})$ to second order about X_t , using $\nabla_X E = g_X$ (established in §5.3) and the Hessian bound of Proposition 7.6. The first-order term recovers $h \dot{E}$; the second-order remainder is bounded by $\frac{h^2}{2} \lambda_{\max}(\nabla^2_X E) \|\dot{X}\|^2$, and $\|\dot{X}\| \leq |F_{\mathrm{base}}| + (1+\gamma)|g_X|$ by the triangle inequality. $\square$

Corollary G1.2 (Step-size for monotone Euler). Energy is non-increasing at the discrete step if

$$h \leq \frac{2 \dot{E}(X_t)}{\Lambda(X_t) \|\dot{X}(X_t)\|^2}.$$

Combined with the adaptive step rule $h_t = h_0 / (1 + \lambda_{\max}(\nabla^2_X E(X_t)))$ from §H.1.4, monotonicity holds whenever $\|\dot{X}\|^2 \leq 2(1 + \Lambda) / \Lambda \cdot (-\dot{E}) / h_0$, which is satisfied on compact sublevel sets by the boundedness of $\dot{X}$ (Theorem 9.4).

Corollary G1.3 (ϵ -regularisation bias). For the regularised ODE with denominator $|g_X|^2 + \epsilon$, the energy derivative satisfies

$$\dot{E}_\epsilon = \dot{E} + \gamma \frac{\epsilon}{|g_X|} \langle F, g_X \rangle \frac{1}{|g_X|^2 (|g_X|^2 + \epsilon)}.$$

The bias term is $O(\epsilon / |g_X|^2)$ for $\epsilon \ll |g_X|^2$, and vanishes as $\epsilon \rightarrow 0$. On sublevel sets where $|g_X|^2 \geq \delta > 0$ the bias is uniformly $O(\epsilon / \delta)$, so the regularised system satisfies $\dot{E}_\epsilon \leq 0$ provided $\epsilon < \delta \cdot |g_X|^2 / |\langle F, g_X \rangle|$. **Practical rule:** choose $\epsilon = 10^{-8} \cdot \text{median}(|g_X|^2)$ from a warm-up run.

G1.2 Runge–Kutta (RK4)

RK4 has local truncation error $O(h^5)$ and global error $O(h^4)$. The energy monotonicity condition weakens to requiring $h < h^*(X_t)$ where h^* is the same bound as Corollary G1.2 but with the coefficient $\frac{1}{2}$ replaced by $\frac{1}{120}$ (from the RK4 error constant). In practice, RK4 with the adaptive step rule of §H.1.4 is unconditionally safe on compact sublevel sets. For stiff problems (large $\lambda_{\max}(\nabla^2_X E)$), implicit midpoint is preferred: it is symplectic and energy-non-increasing by construction.

G2. Stochastic Extension (Open Problem P6)

G2.1 Canonical SDE

Define the canonical SDE

$$dX = H(X)dt + \Sigma(X)dW_t,$$

where $H(X)$ is the canonical-ODE drift (Section 4) and W_t is standard n -dimensional Brownian motion.

Theorem G2.1 (Itô energy equation). Under (A1)–(A4) and $\Sigma \in C^1(\mathbb{R}^n, \mathbb{R}^{n \times m})$ with locally bounded operator norm,

$$dE = \langle g_X, H \rangle dt + \langle g_X, \Sigma dW_t \rangle + \frac{1}{2} \text{tr}(\Sigma^T \nabla^2_X E \Sigma) dt.$$

The expected energy dynamics are

$$\frac{d}{dt} \mathbb{E}[E] = (1 - \gamma) |\langle g_X, F_{\mathrm{base}} \rangle| - |g_X|^2 + \frac{1}{2} \operatorname{tr}(\Sigma^T \nabla^2_X E, \Sigma) \quad (5.3)$$

Proof. Apply Itô's lemma to $E(X_t)$ with $f = E \in C^2$: $dE = (\nabla_X E)^T dX + \frac{1}{2} (dX)^T \nabla^2_X E (dX)$. Since $\nabla_X E = g_X$ (5.3) and $(dX)^T \nabla^2_X E (dX) = \operatorname{tr}(\Sigma^T \nabla^2_X E, \Sigma) dt$ in the Itô sense, the result follows by substituting the canonical ODE for the drift and using Lemma 7.7. $\square$

G2.2 Fisher-optimal noise

Proposition G2.2 (Fisher noise and chart-independence). Under the Fisher choice $G(X) = I_S(S(X))$ and noise covariance $\Sigma(X) = I_X^{\dagger/2}$ (Moore-Penrose square root of the state Fisher metric), the Itô correction on the target set $\{S = 0\}$ reduces to

$$\frac{1}{2} \operatorname{tr}(\Sigma^T \nabla^2_X E, \Sigma) \Big|_{\{S=0\}} = \frac{1}{2} \operatorname{rank}(I_X),$$

which is chart-independent (an integer) and equals the dimension of the identifiable tangent subspace.

Proof. On $\{S=0\}$, $\eta = 2GS = 0$, so $\nabla^2_X E = 2J^T I_S J = 2I_X$ (5.3). Then $\operatorname{tr}(I_X^{\dagger/2} \cdot 2I_X \cdot I_X^{\dagger/2}) = 2 \operatorname{tr}(I_X^{\dagger/2} I_X I_X^{\dagger/2}) = 2 \operatorname{rank}(I_X)$, using the identity $I_X^{\dagger/2} I_X I_X^{\dagger/2} = P_{\mathrm{range}(I_X)}$. Chart-independence follows from Theorem I.1: $\operatorname{rank}(I_X)$ is invariant under the transformation $\tilde{I}_X = T^T I_X T$. $\square$

Corollary G2.3 (Free-energy identification). Define the free energy $\mathcal{F} = \mathbb{E}[E] - \frac{1}{2} \operatorname{rank}(I_X) \cdot t$. Under the Fisher noise, $\mathcal{F}$ is non-increasing along $\frac{d}{dt} \mathbb{E}[E]$ whenever the deterministic descent condition of Theorem 9.1 holds. The Itô correction is therefore a constant-rate drift, not an instability source.

G2.3 Stochastic ultimate boundedness

Theorem G2.4 (Stochastic ultimate boundedness). Let $c_\Sigma := \frac{1}{2} \sup_X \operatorname{tr}(\Sigma^T \nabla^2_X E, \Sigma)$. Under the hypotheses of Theorem 9.4 with admissibility condition $M + c_\Sigma < \Psi^*$, there exist $0 < R_- \leq R_+$ such that $E(X_t)$ is ultimately bounded in $[R_-, R_+]$ almost surely.

Proof. The Itô correction enters as an additional effective disturbance of amplitude c_Σ . Replace M by $M_{\mathrm{eff}} = M + c_\Sigma$ in the proof of Theorem

9.4. The unimodal structure of Ψ is unchanged; the roots R_{pm} shift to absorb the noise term. Since c_{Σ} is bounded on compact sublevel sets (by (A1) and boundedness of Σ), the argument carries through. $\square$

Practical guidance: For BSDT applications, the Itô correction is $c_{\Sigma} = \frac{1}{2} \text{operatorname{tr}}(\Sigma_0^{-1} \cdot \Sigma_{\text{noise}})$ where Σ_0 is the calibration covariance and Σ_{noise} is the measurement noise covariance. The adjusted admissibility threshold is $\Psi_{\text{eff}} = \Psi^* - c_{\Sigma}$.

G3. Time-Varying $G(t)$ and Updated Exponential Rate (Open Problem P7)

Theorem G3.1 (Exponential rate under metric flow). Under the hypotheses of Theorem F.2 with $F_{\text{base}} \equiv 0$, constant G^* , and the relaxation schedule $\dot{G} = \sigma_{\text{rel}}(G^* - G)$, the exponential decay rate of E under the coupled (X, G) dynamics is

$$E(X(t)) \leq E(X_0) e^{-\rho_{\text{flow}} t}, \quad \rho_{\text{flow}} = \rho_F \cdot \frac{\mu(t)}{\mu(t) + \delta_G(t)},$$

where $\rho_F = 4\mu_{\text{I}}^2 \sigma_{\min}^2(J)(1 - \gamma_{\max})/M_{\text{I}}$ is the frozen-metric rate of Theorem J.1, $\mu(t) = \lambda_{\min}(G(t))$, and $\delta_G(t) = |G(t) - G^*|_F / |G^*|_F$ is the relative metric error at time t .

Proof sketch. From Theorem F.2, $V(X, G) = E + \frac{1}{2} \sigma_{\text{rel}}^{-1} |G - G^*|_F^2$ is non-increasing. The metric-error term satisfies $|G(t) - G^*|_F \leq |G(0) - G^*|_F e^{-(\sigma_{\text{rel}} t)}$ (linear relaxation ODE). Substituting into the energy-descent inequality, $\dot{E} \leq -(1 - \gamma) |g_X|^2 \cdot \mu(t) / (\mu(t) + \delta_G(t))$ and applying the Grönwall argument of Theorem 9.3 with the time-varying coefficient gives the displayed rate. $\square$

Remark G3.2 (Rate monotonicity). As $t \rightarrow \infty$, $G(t) \rightarrow G^*$, $\delta_G(t) \rightarrow 0$, and $\rho_{\text{flow}}(t) \rightarrow \rho_F$. The metric-flow system therefore converges to the frozen-metric rate from below. The speed of this convergence is controlled by σ_{rel} : large σ_{rel} collapses the gap rapidly at the cost of larger transient metric oscillations.

Corollary G3.3 (Optimal relaxation rate). The integrated energy bound $\int_0^\infty E(t) dt$ is minimised at $\sigma_{\text{rel}}^* = \rho_F \cdot \mu(0) / |G(0) - G^*|_F$, balancing fast metric convergence against energy overshoot during the transient.

G4. Rank Condition: Quantitative Near-Singularity Bounds

G4.1 The gap

Theorem 9.3 requires $\sigma_{\min}(J(X)) \geq \sigma > 0$ on the compact sublevel set. Near kinematic singularities (robotics) or degenerate calibrations (BSDT), this fails. The regularised ODE ($\epsilon > 0$) is prescribed but without quantifying the convergence degradation.

G4.2 Degraded-rank convergence theorem

Theorem G4.1 (Near-singular convergence). Let $\sigma_{\min}(J(X)) \geq \sigma(X) > 0$ on $\{E \leq c\}$ but allow $\sigma(X) \rightarrow 0$ near the singular set $\mathcal{S}$. Define the effective rate

$$\rho_{\mathrm{eff}}(X) = \frac{4\sigma(X)^2 \mu_G^2}{M_G(1-\gamma(X))}.$$

Then for the regularised ODE ($\epsilon > 0$), $E(X(t)) \leq E(X_0) \exp(-\int_0^t \rho_{\mathrm{eff}}(X(s)) ds)$.

Proof. From the proof of Theorem 9.3, $|g_X|^2 \geq 4\sigma(X)^2 \mu_G^2 / M_G \cdot E$, using the pointwise singular value $\sigma(X)$ in place of the uniform σ . Grönwall's inequality with the time-varying coefficient gives the exponential bound with the integrated rate. $\square$

G4.3 Singularity avoidance condition

Proposition G4.2. The canonical ODE avoids $\mathcal{S}$ along trajectories starting in $\{E \leq c\}$ if

$$\inf_{X \in \{E \leq c\}} \sigma_{\min}(J(X)) \geq \sigma_{\min}^{\mathrm{crit}} := \sqrt{\frac{M_G M}{4\mu_G^2(1-\gamma_{\max})}},$$

where M is the disturbance bound of Theorem 9.4. Below this threshold, the gradient g_X is too weak to prevent F_{base} from driving the trajectory toward $\mathcal{S}$.

Operational check: Before running the canonical ODE on a new domain, compute $\sigma_{\min}(J(X_0))$ and compare to $\sigma_{\min}^{\mathrm{crit}}$. For BSDT with $J = I$ (affine SS), this condition is automatic ($\sigma_{\min} = 1$). For task-space robotics, compute the minimum manipulability $w(q) = \sqrt{\det(J(q)J(q)^T)}$ over the trajectory and flag if $w < \sigma_{\min}^{\mathrm{crit}}$.

G5. False-Positive Rate: ERCOT Early-Warning Validation Protocol

G5.1 The gap

The ERCOT result (+1033 h lead from hour 23) gives no false-positive denominator. Without a false-positive rate, the lead time statistic is uninterpretable as a detection result.

G5.2 Proposed validation protocol

Definition G5.1 (Canonical alarm). An alarm fires at time t when $E(t) > \mu_{1:t-1} + 2\sigma_{1:t-1}$ for $t \geq 20$ (as defined in §1.4 of the report). A **true positive** (TP) is an alarm within $[-L_{\max}, 0]$ steps of a labelled crisis onset. A **false positive** (FP) is any alarm outside all crisis windows.

Protocol G5.2 (ERCOT FPR estimation):

1. Run the canonical alarm on the full 2019–2022 ERCOT series ($T = 8,760$ h), using only non-crisis hours to define the background distribution.
2. Label crisis windows: Winter Uri (hours 1,040–1,200), plus any NERC-declared emergency events in the data.
3. Count FP alarms per 1,000 non-crisis hours. Denote this λ_{FP} .
4. Compute the detection precision at lead L : $\mathrm{Precision}(L) = \mathrm{TP}(L) / (\mathrm{TP}(L) + \mathrm{FP}(L))$.
5. Report the lead-precision frontier: the curve $(\mathrm{lead}, \mathrm{precision})$ for $L \in [0, 1033]$ h.

Proposed benchmark thresholds for ERCOT:

- $\lambda_{\mathrm{FP}} < 5$ per 1,000 h (i.e., $< 0.5\%$ false alarm rate) is deployment-grade for grid operators.
- $\lambda_{\mathrm{FP}} < 20$ per 1,000 h is research-grade, consistent with existing grid stress literature.

Expected result: The BSDT camouflage channel (δ_C) alarm at hour 23 is from solar volatility exceeding its 2σ envelope — a condition likely to recur during routine high-variability periods (summer afternoons, storm fronts). The FPR is expected to be non-negligible. The question is whether it falls within the benchmark range.

G5.3 Generalised FPR formula

For any domain with background rate λ_{bg} and alarm threshold $\mu + k\sigma$:

$$\lambda_{\mathrm{FP}} = \lambda_{\mathrm{bg}} \cdot 2(1 - \Phi(k)),$$

where Φ is the standard normal CDF. For $k = 2$ (as used in the report), $\lambda_{\mathrm{FP}} \approx 0.046 \cdot \lambda_{\mathrm{bg}}$. This means roughly 1 false alarm per 22 time steps under normality. The expanding-window estimator of μ and σ reduces the effective false-positive rate below this floor, but by how much depends on the autocorrelation of $E(t)$ — which must be measured per domain.

G6. Fisher Tracking Schedule: Practitioner Guidance on Chart Dependence

G6.1 The gap

Remark I.2 states that the locked canonical ODE is chart-dependent under the tracking schedule (F1). No guidance is given on when this matters in practice or how to detect it.

G6.2 Chart-dependence severity metric

Definition G6.1 (Reparametrisation sensitivity). For a diffeomorphism $X = \Phi(\tilde{X})$ with Jacobian $T = \partial\Phi/\partial\tilde{X}$, the reparametrisation sensitivity of the canonical trajectory is

$$\kappa_{\mathrm{rep}}(X) := \frac{|T^{-1}H(X) - H(\Phi^{-1}(X))|}{|H(X)|},$$

where $H(X)$ is the canonical-ODE drift. For the Euclidean canonical ODE, $\kappa_{\mathrm{rep}} = \|I - T^{-1}\|_{\mathrm{op}} + O(\|T - I\|^2)$ near the identity.

Proposition G6.2. Chart dependence is negligible (i.e., $\kappa_{\mathrm{rep}} < \delta$ for a tolerance δ) whenever $\|T - I\|_{\mathrm{op}} < \delta/2$, i.e., the Jacobian of the reparametrisation is close to the identity. This holds whenever the domain coordinates are approximately Cartesian in the operating region.

G6.3 Decision rule for practitioners

Condition	Recommendation
J is affine ($J = \text{const}$)	Use locked ODE. Chart dependence is absent: Euclidean and Fisher projections coincide when J is constant.

S is nonlinear, operating region has $\kappa(I_X) < 10$	Use locked ODE with tracking schedule (F1). Chart dependence is mild.
S is nonlinear, $\kappa(I_X) > 10$ in operating region	Use Fisher-affine coordinates: reparametrise $\tilde{X} = I_X^{1/2} X$ locally. Euclidean projection in $\tilde{X}$ is nearly optimal.
Chart covariance is required (e.g., theoretical guarantees needed across parametrisations)	Use the natural-gradient variant $\dot{X} = -g^f_X$. Accept that Rule 2 is broken. Document this explicitly.

BSDT-specific note: For BSDT with $S(X) = \operatorname{vec}(\tilde{X})$ (affine), $J = I$ and $\kappa_{\mathrm{rep}} = 0$ identically. Chart dependence is a non-issue for the core BSDT calibration. It becomes relevant only when extending to nonlinear feature maps (e.g., deep-feature BSDT) or when the state space undergoes a basis change between calibration and execution.

G7. Admissibility Threshold Sensitivity to Calibration Conditioning

G7.1 The gap

The admissibility threshold $\Psi^* = \sigma_{\theta} \mu_G / \sqrt{M_G}$ depends on μ_G and M_G , the eigenvalue bounds of G . For BSDT, $G = \Sigma_0^{-1}$, so $\mu_G = 1/\lambda_{\max}(\Sigma_0)$ and $M_G = 1/\lambda_{\min}(\Sigma_0)$. The dependence on $\kappa(\Sigma_0) = \lambda_{\max}/\lambda_{\min}$ is not explicit.

G7.2 Explicit conditioning bound

Proposition G7.1 (BSDT admissibility and calibration conditioning). For $G = \Sigma_0^{-1}$,

$$\Psi^* = \frac{\sigma_{\theta}}{\lambda_{\max}(\Sigma_0) \sqrt{1/\lambda_{\min}(\Sigma_0)}} = \frac{\sigma_{\theta} \sqrt{\lambda_{\min}(\Sigma_0)}}{\lambda_{\max}(\Sigma_0)} = \frac{\sigma_{\theta}}{\kappa(\Sigma_0)^{1/2} \lambda_{\max}^{1/2}(\Sigma_0)}.$$

Equivalently, $\Psi^* = \sigma_{\theta} / \sqrt{\kappa(\Sigma_0) \lambda_{\max}(\Sigma_0)}$.

Corollary G7.2. The admissibility threshold degrades as $1/\sqrt{\kappa(\Sigma_0)}$. A $10\times$ increase in calibration conditioning shrinks the safe disturbance budget by $\sqrt{10}$

$\approx 3.16 \times$.

G7.3 Calibration conditioning guidelines

Practical bounds for BSDT:

$\kappa(\Sigma_0)$	$\Psi^*/\Psi^*_{\kappa=1}$	Status
< 10	> 0.32	Safe: standard calibration
$10 - 100$	$0.10 - 0.32$	Warning: consider regularisation
> 100	< 0.10	Critical: Ψ^* is $< 10\%$ of ideal; mandatory regularisation

Regularisation prescription: Replace Σ_0 by $\Sigma_0^{\lambda} = \Sigma_0 + \lambda I$ with

$\lambda^* = \arg\min_{\lambda} \left[\kappa(\Sigma_0 + \lambda I) \right] \text{ subject to } \kappa(\Sigma_0^{\lambda}) \leq \kappa_{\max}.$

For a target $\kappa_{\max} = 10$: $\lambda^* = (\lambda_{\max}(\Sigma_0) - 10, \lambda_{\min}(\Sigma_0))/9$ when $\kappa(\Sigma_0) > 10$; $\lambda^* = 0$ otherwise.

Effect on θ : The control parameter θ should be re-calibrated after regularisation as $\theta^{\text{new}} = \theta \cdot \kappa(\Sigma_0)/\kappa_{\max}$ to hold Ψ^* constant.

Summary: Gap Status

Gap	Open problem	Resolution	Status
P5: Discretisation error	Yes	Prop. G1.1–G1.3: Euler bound, RK4 note, ε -bias formula	Closed
P6: Stochastic extension	Yes	Thm. G2.1–G2.4: Itô equation, Fisher noise, stoch. UB	Closed
P7: Time-varying G rate	Yes	Thm. G3.1–Cor. G3.3: ρ_{flow} , monotonicity,	Closed

		optimal σ_{rel}	
Rank condition near singularities	No (practitioner gap)	Thm. G4.1–Prop. G4.2: degraded-rank rate, operational check	Closed
ERCOT false-positive rate	No (empirical gap)	Protocol G5.2–G5.3: FPR formula, benchmark thresholds	Protocol provided
Fisher tracking chart dependence	No (practitioner gap)	Def. G6.1–G6.3: severity metric, decision table	Closed
Admissibility vs. $\kappa(\Sigma_0)$	No (calibration gap)	Prop. G7.1–G7.3: explicit bound, regularisation prescription	Closed

This addendum is logically downstream of the locked core (§13 of v4). No canonical object (g_X, F, γ , the ODE) is redefined. Results in G1–G3 resolve the open problems listed in §C of v4. Results in G4–G7 close practitioner gaps identified in external structural review.